SYNOPSIS

Project Title

Two-Player Console-Based Tic-Tac-Toe Game

Objective

Design and build an interactive 3X3 grid game in Python demonstrating core programming fundamentals such as lists, loops, conditional statements, functions, and input validation without external libraries.

Key Features:

- **Game Board:** Represented using a single list of 9 elements, displaying a neatly formatted console grid after every move.
- **Two-Player Turns:** Alternates turns between Player 1 and Player 2 dynamically using their custom names.
- **Input Validation:** Handles non-numeric inputs, out-of-range numbers (outside 1–9), and already occupied cells gracefully without crashing.
- **Win & Draw Detection:** Automatically checks all 3 rows (horizontal), 3 columns (vertical), and 2 diagonals after each move to see if a player has three marks in a line.
- **Draw Detection:** Detects when all 9 cells are filled with no winner, ending the game in a draw.
- **Replay Option:** Allows users to play multiple rounds or exit with a closing message.

SANDRA BOSE